

World

Robotics

Service Robots

2026

incl. Mobile and Medical Robots



**Statistics, Market Analysis, List of Manufacturers and
Case Studies**

World Robotics Service Robots

2026

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World Robotics 2026 – Service Robots

Produced by VDMA Services GmbH, Lyoner Str. 18, 60528 Frankfurt, Germany.

The service robot statistics is carried out by the IFR Statistical Department.

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The cover and the editorial are sponsored by Innok Robotics.

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Suggested citation: Jurkat, Anne; Müller, Christopher; Kraus, Werner, Graf, Birgit; Bregler, Kevin (Eds.): World Robotics 2026 – Service Robots, IFR Statistical Department, VDMA Services GmbH, Frankfurt am Main, Germany, 2026.

Short citation: World Robotics 2026 – Service Robots

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ISBN 978-3-8163-0779-2

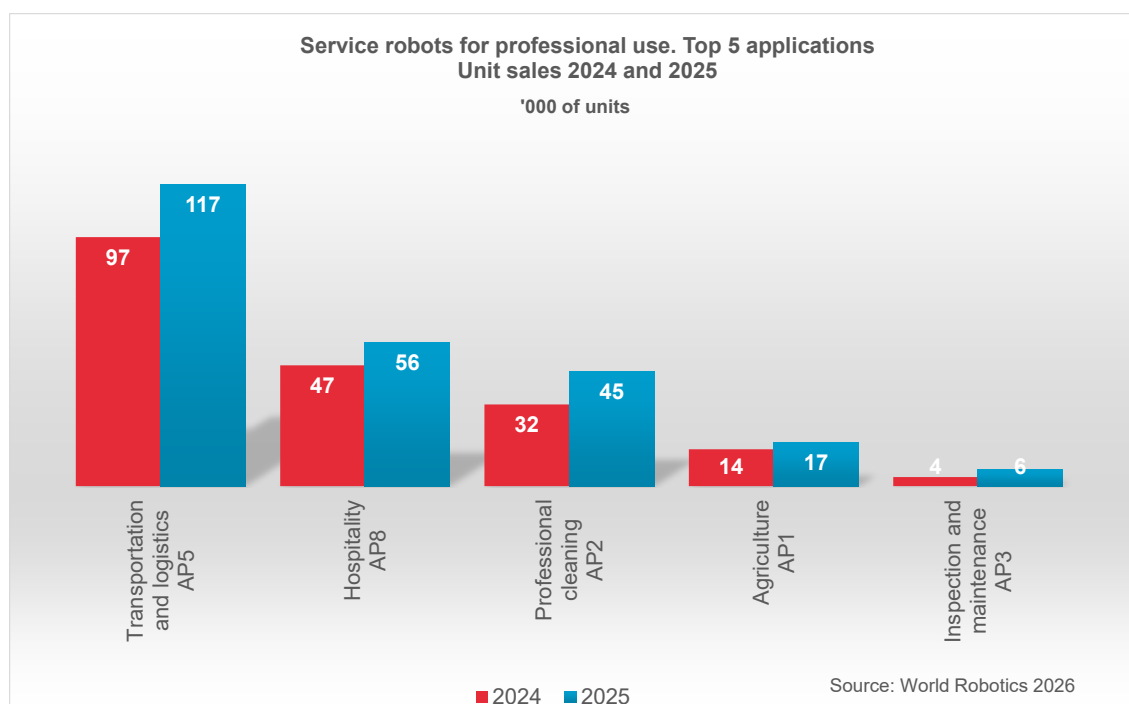
Executive Summary World Robotics 2026 - Service Robots

The service robot industry continues to be highly diverse, with applications ranging from logistics and professional cleaning to agriculture, healthcare, hospitality, and emerging humanoid robots. The IFR Statistical Department is currently aware of more than 1,000 service robot producers worldwide. This excludes integrators and prototyping services.

In 2025, worldwide sales of professional service robots (excluding medical robots) increased by 24%. At the same time, Robotics-as-a-Service (RaaS) continued to gain importance with a high double-digit growth rate. Economic uncertainty, high upfront investment costs, and the need for greater operational flexibility are supporting the adoption of subscription- and usage-based business models. RaaS makes automation more accessible, particularly for small and medium-sized enterprises, while allowing companies to scale robotic operations according to demand.

Top applications for professional service robots

The five largest application groups for professional service robots in 2025 were transportation and logistics (AP5), hospitality (AP8), professional cleaning (AP2), agriculture (AP1), and inspection and maintenance (AP3).



Transportation and logistics remained by far the largest professional application group. Nearly every second professional service robot sold in 2025 was built for the transportation of goods or cargo. Mobile robot-based order fulfillment solutions are accelerating installations worldwide, while emerging applications such as truck loading

and unloading, outdoor transportation, and cleanroom logistics provide additional opportunities for future growth.

Demand for **hospitality robots** (AP8) also increased strongly. Mobile guidance, information, and telepresence robots continue to account for most sales, while food and drink preparation is emerging as a promising new field. Labor shortages, rising operating costs, and increasing demand for consistent food production are expected to accelerate robot adoption in corporate cafeterias, hospitals, airports, convenience retail, quick-service restaurants, and other professional environments. Social interaction robots for professional use are also becoming increasingly viable as advances in generative AI and large language models (LLMs) improve their capabilities.

Demand for **professional cleaning robots** (AP2) expanded particularly strongly. Growth was driven mainly by floor cleaning robots, reflecting the strong potential for automation and scalability in this application. At the same time, new products are entering the market, including outdoor sweepers and professional robotic lawn mowers. These solutions are particularly attractive for municipalities seeking to maintain parks and public spaces while facing a shrinking workforce. Declining product prices and the increasing availability of RaaS models are expected to further lower adoption barriers.

Agricultural robots (AP1) also returned to solid growth in 2025. Labor shortages and climate change continue to be major drivers of automation, while advances in intelligent vision systems are expected to accelerate the adoption of robots for precision farming. Current growth is increasingly supported by barn cleaning and feeding robots in addition to the established markets for cultivation and milking robots. High upfront capital expenditure, difficult outdoor environments, and gaps in rural connectivity nevertheless remain important barriers to broader adoption.

The market for **inspection and maintenance robots** (AP3) recorded another year of exceptionally strong expansion. Recent advances in artificial intelligence, sensor technology, and autonomous navigation are enabling robots to perform inspection tasks in hazardous and hard-to-access environments with increasing autonomy. The need to maintain and modernize critical infrastructure is expected to continue driving demand in this market.

Medical robots

The market for **medical robots** continued to offer attractive growth opportunities in 2025. Surgical robotics was a particularly important driver, while exoskeletons also recorded a substantial increase in unit sales. Demographic change is expected to remain one of the most important long-term market drivers. Population aging simultaneously reduces the availability of healthcare personnel and increases demand for medical treatment and support. Robotics applications are therefore expanding beyond traditional surgical systems into laboratory automation, rehabilitation, elderly care, and other healthcare-related services.

Consumer use robots

Service robots for **consumer use** continued to represent a mass market and recorded strong growth in 2025. Robots for domestic tasks remain by far the largest category, led by robotic vacuum cleaners. Window cleaning robots also developed dynamically, while robotic lawn mowers and pool cleaning robots serve smaller but well-established consumer markets. Advances in functionality are supporting demand, although cybersecurity requirements, trade restrictions, sustainability considerations, and increasing product complexity are becoming important challenges for manufacturers.

Artificial intelligence is also revitalizing the market for **social interaction robots**. While many earlier social robots struggled to achieve widespread adoption because of high costs and limited functionality, recent progress in generative AI and especially large language models is creating new opportunities for AI-powered social companions. The education robot market remains promising as parents and teachers worldwide place increasing emphasis on STEM education.

Humanoid robots

A major new area covered by World Robotics is the emerging market for **humanoid robots**. Almost 6,900 humanoids were reported sold globally in 2025, with the overwhelming majority originating from companies headquartered in Asia. The market is still at a very early stage: many companies offer humanoids primarily for testing, research, and pilot projects rather than for broad commercial deployment. The IFR Statistical Department is currently aware of 171 humanoid robot manufacturers, illustrating the considerable entrepreneurial activity surrounding this technology.

Humanoid robotics is also one of the youngest parts of the robotics industry. A large share of identified manufacturers was established only in recent years, with a pronounced wave of new company formation emerging alongside advances in generative artificial intelligence. Whether all of today's developers will achieve market readiness remains uncertain, making humanoids one of the most dynamic fields to watch over the coming years.

Outlook

Across service robotics, the underlying drivers remain powerful. Demographic change, persistent labor shortages, rising labor costs, advances in artificial intelligence and sensor technology, and the need to improve productivity are creating opportunities for automation across an expanding range of applications. At the same time, geopolitical uncertainty, trade restrictions, cybersecurity requirements, regulation, and high investment costs can slow down adoption. Flexible business models such as RaaS can help overcome some of these barriers and are expected to gain further importance.

The service robotics market therefore combines established high-volume applications with a growing number of specialized and emerging solutions. Some application groups have already reached considerable scale, while others are only beginning to transition from technological development to commercial deployment. World Robotics Service Robots 2026 provides detailed statistics by application, region, and business model, complemented by market assessments and outlooks for the major segments. The report also provides an in-depth analysis of the global service robotics industry and the companies shaping its future.

1 Introduction

Chapter 1 reviews definitions and classifications of service robots.

1.1 DEFINITIONS: ROBOTICS, SERVICE ROBOTICS, INDUSTRIAL ROBOTICS, AND MEDICAL ROBOTICS

1.1.1 ISO 8373:2021 VOCABULARY DEFINITIONS

ISO 8373:2021 “*defines terms used in relation to robotics*” (§ 1). These vocabulary definitions relate to industrial robotics, service robotics, and medical robotics. This section describes the ISO definitions needed to understand IFR classification schemes and to distinguish IFR industrial robot statistics and IFR service robot statistics.

According to ISO 8373:2021, a **robot** is a “*programmed actuated mechanism with a degree of autonomy, to perform locomotion, manipulation or positioning*” (§ 3.1). Mechanisms that are using robot technology but do not satisfy the definition of robot, for instance teleoperated manipulators, are called **robotic device**. (§ 3.5). Robot technology includes perception, reasoning and planning algorithms (§ 3.3).

A **service robot** is a robot “*that performs useful tasks for humans or equipment*” (§ 3.7).

A **medical robot** is a “*robot intended to be used as medical electrical equipment*” (§ 3.8).

An **industrial robot** is an “*automatically controlled, reprogrammable multipurpose manipulator, programmable in three or more axes, which can be either fixed in place or fixed to a mobile platform for use in automation applications in an industrial environment*” (§ 3.6). Chapter 1.7 of World Robotics Industrial Robots provides an in-depth elaboration of IFR’s definition of industrial robots and industrial robot classification schemes.

A **mobile robot** is by § 4.15 a robot that can “*travel under its own control.*” By § 4.16, automated guided vehicles (AGV) are mobile platforms “*following a predetermined path indicated by markers or external guidance commands...*” An AGV is therefore not a robot but a robotic device as it lacks the autonomy to determine its own path or navigate without external guidance. Mobile robots can travel on wheels (§ 4.15.1), legs (§ 4.15.2 and § 4.15.3) or by crawling on caterpillar tracks (§ 4.15.4). Note that mobility per se does not assign a robot to a specific category (industrial, service, medical). Note further that § 4.15 does not mention swimming, diving, or flying as types of locomotion. But from the example of underwater inspection provided in § 6.17 (teleoperation), it follows that at least diving is a type of mobility that is not ruled out generally.

A wearable robot (§ 4.17) is a robot “*attached to and carried by a human during use*”, providing “*assistive force*”. In practice, this describes powered exoskeletons and powered prostheses.

ISO 8373:2021 further distinguishes personal from professional service robots (§ 3.7) but does not provide any definition of these terms. There are just a few examples of tasks considered as personal use (note 1 to § 3.7) and tasks considered as professional use (note 2 to § 3.7). The examples in both notes intersect to a large extent. Tasks like “*handling of items*”, “*providing guidance or information*”, “*cooking and food handling*”, and “*cleaning*” are named as examples for personal as well as for professional use.

1.1.2 DEVIATIONS OF IFR DEFINITIONS FROM ISO DEFINITIONS AND IFR REFINEMENTS OF ISO DEFINITIONS – SERVICE ROBOTS AND MEDICAL ROBOTS

IFR generally defines robots according to ISO 8373:2021. There are, however, some details that are by IFR's experience not helpful to unambiguously distinguish the different robot categories, or that might contrast with the primary goal of IFR statistics – which is to provide information on the robotics industry to inform the robotics industry. IFR service and medical robot statistics will therefore deviate from ISO definitions in specific details described in this section.

Delimitation of industrial robots, service robots and medical robots

The ISO definition of “industrial robot” creates some room for interpretation. First, it is not clearly defined what constitutes an industrial environment and what separates it from other types of environments, e.g. workshops or laboratories. Second, as the previous definition by application (see ISO 8373:2012) was abandoned only for industrial robots, but not for service robots and the newly created category of medical robots, there is some inconsistency in the definition. Previously, the definition of the term service robot included that it is not used in “*industrial automation applications*” (§ 2.10, ISO 8373:2012). Now, this part of the service robot definition is deleted. This means that service robots can be used in industrial automation, but they still do not qualify as industrial robots if they do not satisfy the additional requirements listed in § 3.6.

The definition of **service robots by application** has proven successful in the history of IFR statistics. It also does not require a definition of the environment the robot is used in. The IFR application classification schemes for service and medical robots (see chapter 1.4), developed by the IFR Robot Supplier Committee and IFR Service Robot Committee, define various service and medical robot applications. Note: The Robot Supplier Committee decided to include all robots of typically industrial kinematics in the industrial robot statistics. Robots with industrial robot kinematics used in service applications are, therefore, counted in both statistics: Since 2024, the IFR industrial robot statistics count such cases in application classes 904 (“service applications”) and 905 (“medical applications”). Up to 2023, such cases were included in application class 909 (“other applications”). All robots reported to these classes shall also be reported to IFR service robot statistics.

Medical robots used to be an application group in the professional service robot application scheme. Starting in World Robotics 2024, IFR statistics follow the new ISO definition and upgrade medical robots from an application group to a robot category. The classification in statistics, however, does not change and the old application group label AP6 is kept for medical robots in IFR service robot statistics.

1.1.4 SUMMARY: IFR SERVICE AND MEDICAL ROBOT DEFINITION

- A **service robot** is a programmed actuated mechanism with a degree of autonomy to perform locomotion, manipulation or positioning to perform useful tasks for humans or equipment.
- In some applications, manually operated **robotic devices** with limited or even without autonomy are included.
- A **consumer service robot** is a service robot built for use by everyone. Neither safe operation nor setup require a professionally trained operator.
- A **professional service robot** is a service robot built for use by trained professional operators.
- **Autonomous mobile robot (AMR)** is a marketing term that is usually used as a synonym for wheeled mobile robots used in professional applications.
- A **medical robot** is intended to be used as medical electrical equipment.

1.2 COMPLIANCE AND PRIVACY

IFR Statistical Department ensures confidentiality of individual company data and compliance with antitrust regulations. Access to raw data is strictly limited to the IFR Statistical Department staff. The IFR Statistical Department will never provide company-level data to third parties neither outside nor inside the IFR. The IFR Statistical Department publishes only aggregated data. The IFR Statistical Department will not reveal data if a data point consists of less than four observations. This is to prevent mathematical retrieval of company-level data.

1.3 CLASSIFICATION OF SERVICE AND MEDICAL ROBOTS

The IFR is using two schemes to classify service and medical robots: the application and the type of movement. Both classification schemes consist of classes at the lowest level, which are aggregated into groups.

There are four major types of movement that serve as the groups of this scheme: Ground-based (A), water-based (B), aerial (C), and wearable (D). The fifth group, type E is a residual group that covers everything not covered by A to D. On the class-level, ground-based robots can either be rolling (A1), walking (A2), be fixed in place (A3), or have any other ground-based type of movement (A4). Water-based robots can either be swimming (B1) or diving (B2). Aerial robots are usually flying (C1) but there might also be hovering robots in the future (C2). Similarly, wearable robots are powered exoskeletons (D1) today, but there might be other types of wearable robots (D2) in the future. Robots that do not fit into any of these classes, e.g. robots for orbital space or hybrid robots that fit into more than one of the above classes, can be classified as type E1. The full classification scheme including descriptions of each class is presented in Table 1.1.

Table 1.1

Classification of service robots by type of movement

Type		Description
A	Ground-based	Robots that move or stand on the ground
A1	Rolling	Rolling on wheels or caterpillar tracks
A2	Walking	Walking on legs
A3	Fixed in place	Immobile, cannot change physical location by itself, standing on the ground, desk or other fixed place, also hanging
A4	Other ground-based	Ground-based but none of the above (A1-A3), e.g. crawling, snakeing, climbing
B	Water-based	Robots that swim or dive
B1	Swimming	Swim on the surface of the water, Note: If the robot can both swim and dive, it is counted as diving (B2)
B2	Diving	Dive under the surface of the water
C	Aerial	Robots that move through the air
C1	Fly	Flying in the air
C2	Hover	Hover above ground
D	Wearables	Robots that are worn by people
D1	Exoskeletons	Powered human exoskeletons
D2	Other wearables	Wearable robots other than D1
E	Others	Robots that are not A-D
E1	Other robots	Robots that do not fit into classes A-D, e.g. robots for orbital space Robots that fit into multiple classes, e.g. hybrid robots for water and ground or air

Source: IFR

The classification of service robots by application follows the concept outlined in chapter 1.2.4. This means, there are two distinct categories: consumer robots (AC – “applications, consumer”) and professional service robots (AP – “applications, professional”). Chapters 3 and 4 of this book provides comprehensive explanations and examples of robots for each application class.

1.4 SAMPLE DESCRIPTION

The International Federation of Robotics Statistical Department (IFR SD) conducts an annual survey among service robot suppliers worldwide. Data was either sent to the IFR SD directly or through national robotics associations.² Additional data was acquired by comprehensive desktop research using sources such as annual reports and other publicly available information.

The service robot industry is more diverse and less tangible than the industrial robot industry. The IFR SD is currently aware of 1001 service robot producers worldwide. This excludes prototyping services (i.e. companies that develop service robot prototypes upon request but do not intend to go into serial or mass production) and system integrators (i.e. companies that buy a third-party robot to create a service robot application). Many companies are still in the product development stage and do not have marketable products yet. The fact that 22 companies in our sample do not have any sales yet emphasizes this. These companies are still at the funding or prototyping stage and intend to offer a marketable product in the future. IFR Statistical Department is continuously seeking new service robot producers, so please contact statistics@ifr.org if you represent such a company and would like to participate in the annual survey.

The data reported here is sample data. It is not projected to the whole industry. It thus underestimates the actual sales figures and should rather be interpreted as a **minimum level of sales**. As the survey participation and the desktop research generate different sample compositions in each survey wave, the **statistics should be interpreted as cross-sectional**. **The IFR SD strongly discourages the creation of time-series data, compiling data from different issues of this publication.**

Table 1.5 presents more details on the sample and the relation to the statistical population. The statistical population consists of 1001 companies. Thereof 413 companies are from Europe (41% of the total population), 305 companies are from the Asia-Pacific region (35% of the total population), 235 companies are from the Americas (24% of the total population, almost exclusively from North America) and 3 companies are from Africa. The IFR SD is currently not aware of any service robot producer in other regions of the world. Most companies (749; 75% of the population) offer professional service robots and 203 companies (20% of the population) offer consumer service robots. Medical robots are in the portfolio of 137 companies, thereof 39% (53 companies) from Europe, 34% (47 companies) from Asia+Pacific, and 26% (35 companies) from the Americas (in fact: from the United States or Canada). There are 4 companies active in all three segments (consumer, professional, medical) and 59 companies are active in consumer and professional service robotics simultaneously.

The sample includes data from 238 service robot suppliers. Thereof 22 companies reported (or desktop research suggested) that there are no sales yet. These zero reports do not contribute to the minimum number of observations required to satisfy the IFR's

² IFR SD gratefully appreciates the support of AER, CRIA, DIRA, JARA, and KAR.

antitrust compliance rules (see chapter 1.3). The sample includes 89 suppliers from Europe, thereof 12 zero reports, 92 suppliers from the Asia-Pacific region, thereof 2 zero reports, and 57 North American companies, thereof 8 zero reports.

Table 1.5
Service robot suppliers by region of origin: population versus sample

		Total	Consumer (AC)	Professional (AP1-5,7-9)	Medical (AP6)
		companies	companies	companies	companies
Europe	population	413	62	326	53
	sample	89 (12)	11	74 (10)	14 (1)
The Americas	population	235	50	159	35
	sample	57 (8)	4	37 (6)	16 (3)
Asia + Pacific	population	350	91	262	47
	sample	92 (2)	17	71 (2)	13
Africa	population	3	0	2	2
	sample	0	0	0	0
Total	population	1001	203	749	137
	sample	238 (22)	32	182 (18)	43 (4)

Source: World Robotics 2026

The sum of categories AC, AP, and AP6 exceeds the total because a company can be assigned to more than one category.

Population of service robot manufacturers, excluding companies that only do prototyping.

Sample including 22 companies that do not have sales yet (in parentheses).

The sample composition is well balanced in terms of the geographical distribution of the population (see Table 1.6). Across all regions, the share of companies in the sample typically deviates from the corresponding share in the population by no more than 4 percentage points. This also holds for the professional-use category.

There are, however, a few exceptions. In the medical robots category, the Americas are somewhat overrepresented, accounting for 37% of the sample compared with 26% of the population, while Europe is slightly underrepresented (33% of the sample versus 39% of the population). In the consumer category, the Americas are underrepresented (13% of the sample versus 25% of the population), whereas Asia and the Pacific are somewhat overrepresented (53% versus 45%).

Table 1.6
Representativity of the sample by region

		Total	Consumer (AC)	Professional (AP1-5,7-9)	Medical (AP6)
		percent	percent	percent	percent
Europe	Share of population	41%	31%	44%	39%
	Share of Sample	37%	34%	41%	33%
The Americas	Share of population	23%	25%	21%	26%
	Share of Sample	24%	13%	20%	37%
Asia + Pacific	Share of population	35%	45%	35%	34%
	Share of Sample	39%	53%	39%	30%
Africa	Share of population	0%	0%	0%	1%
	Share of Sample	0%	0%	0%	0%
Total	Share of population	100%	100%	100%	100%
	Share of Sample	100%	100%	100%	100%

Source: World Robotics 2026

The sum of categories AC, AP, and AP6 exceeds the total because a company can be assigned to more than one category.

Population of service robot manufacturers, excluding companies that only do prototyping.

Sample including 22 companies that do not have sales yet (in parentheses).

Statistics use the number of robots (“units”) as a measure (see chapter 1.4).

“Robots-as-a-Service” (RaaS) business models are accounted for by the number of robots constituting the “RaaS fleet”, i.e. robots that are available for service. IFR SD defines all business models as RaaS that have the property of the robot hardware remaining with the robot supplier. This includes leasing, hiring, and others and should be seen in contrast to traditional sales, which transfer the property of the robot hardware to the customer.

In June and July 2026, the IFR conducted a separate survey for the forecast. The current forecast is based on these results and on the IFR’s assessment of the service robotics market